

Edge Computing: Benefits for Businesses and Users



Edge computing is not a new term. It has been around since the 1990s, when a group of researchers from the Massachusetts Institute of Technology (MIT) invented a content delivery network (CDN) that could serve web and video content from edge servers deployed close to users, to save bandwidth. This prototype, developed by the MIT spin-off company Akamai, is considered one of the earliest examples of edge computing.

Since then, there has been a lot of discussion around edge computing and the Internet of Things (IoT). However, the technology has really come of age now, thanks mainly to the rising interest in artificial intelligence (AI) and, of course, a much-needed fillip from fast 5G networks and powerful mobile processors.

Edge computing is basically a distributed computing framework that moves processing and storage to the edge of the network—that is, as close as possible to the users and sources of data, such as the Internet of Things (IoT) or local edge servers. So, why would someone want to do that? Because, speed matters! Today, everyone wants instant answers and solutions, and the cloud alone cannot handle this crowd. This is true not just for futuristic or high-stake applications like self-driving cars and military operations but also for common ones like shopping and marketing campaigns.

Delays are costly

From industries to automobiles and traffic signals, every device is racing

to become smart. Relying solely on the cloud to accommodate this exponential growth of users would result in increased latency—an intolerable outcome. And, it can be more serious than a customer getting irritated by a delay in connecting to a local restaurant's menu. A millisecond's delay in decision-making in an autonomous car could cause an accident, or lack of connectivity in a surveillance drone could mean not spotting an impending missile attack! In today's scenario, reducing latency is of utmost importance. On-device processing, in tandem with the cloud and on-premise data centres, can handle this situation very well. It also brings with it other perks like creating personalised customer experiences and improved security.